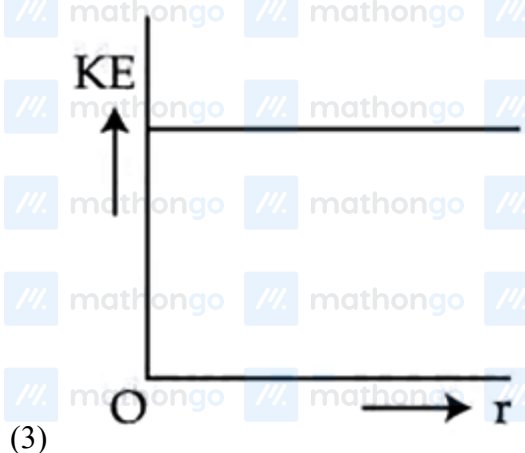
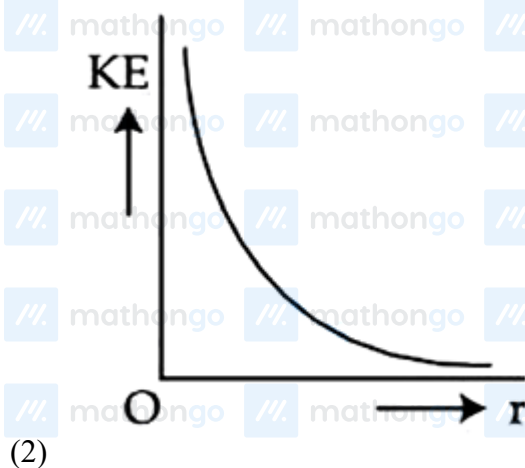
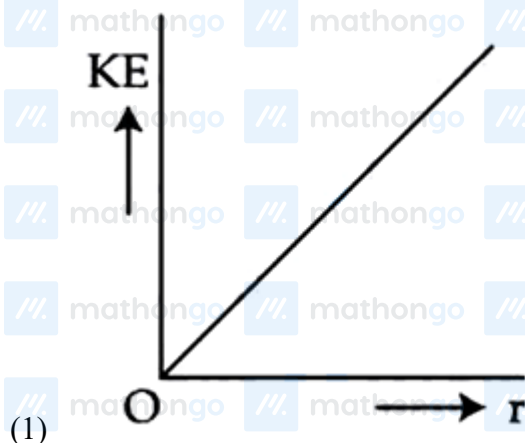


Questions

MathonGo

Q1 - 2024 (04 Apr Shift 1)

An infinitely long positively charged straight thread has a linear charge density $\lambda \text{ Cm}^{-1}$. An electron revolves along a circular path having axis along the length of the wire. The graph that correctly represents the variation of the kinetic energy of electron as a function of radius of circular path from the wire is :



Questions

MathonGo



Q2 - 2024 (04 Apr Shift 1)

An infinite plane sheet of charge having uniform surface charge density $+\sigma_s \text{ C/m}^2$ is placed on $x - y$ plane.

Another infinitely long line charge having uniform linear charge density $+\lambda_e \text{ C/m}$ is placed at $z = 4 \text{ m}$ plane

and parallel to y -axis. If the magnitude values $|\sigma_s| = 2 |\lambda_e|$ then at point $(0, 0, 2)$, the ratio of magnitudes of electric field values due to sheet charge to that of line charge is $\pi\sqrt{n} : 1$. The value of n is _____.

Q3 - 2024 (04 Apr Shift 2)

A charge q is placed at the center of one of the surface of a cube. The flux linked with the cube is:

(1) $\frac{q}{2\epsilon_0}$

(2) $\frac{q}{8\epsilon_0}$

(3) Zero

(4) $\frac{q}{4\epsilon_0}$

Q4 - 2024 (05 Apr Shift 2)

The vehicles carrying inflammable fluids usually have metallic chains touching the ground :

(1) To protect tyres from catching dirt from ground

(2) To alert other vehicles

(3) It is a custom

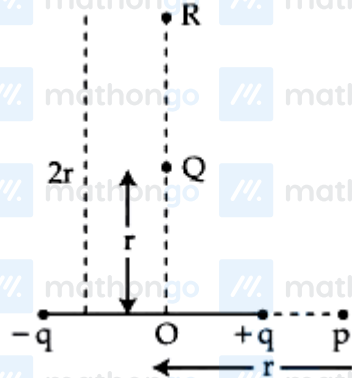
(4) To conduct excess charge due to air friction to ground and prevent sparking

Questions

MathonGo

Q5 - 2024 (05 Apr Shift 2)

The electric field at point p due to an electric dipole is E . The electric field at point R on equatorial line will be $\frac{E}{x}$. The value of x :



Q6 - 2024 (06 Apr Shift 1)

Ques:

σ is the uniform surface charge density of a thin spherical shell of radius R . The electric field at any point on the surface of the spherical shell is :

(1) $\sigma/\epsilon_o R$

(2) σ/ϵ_o

(3) $\sigma/4\epsilon_o$

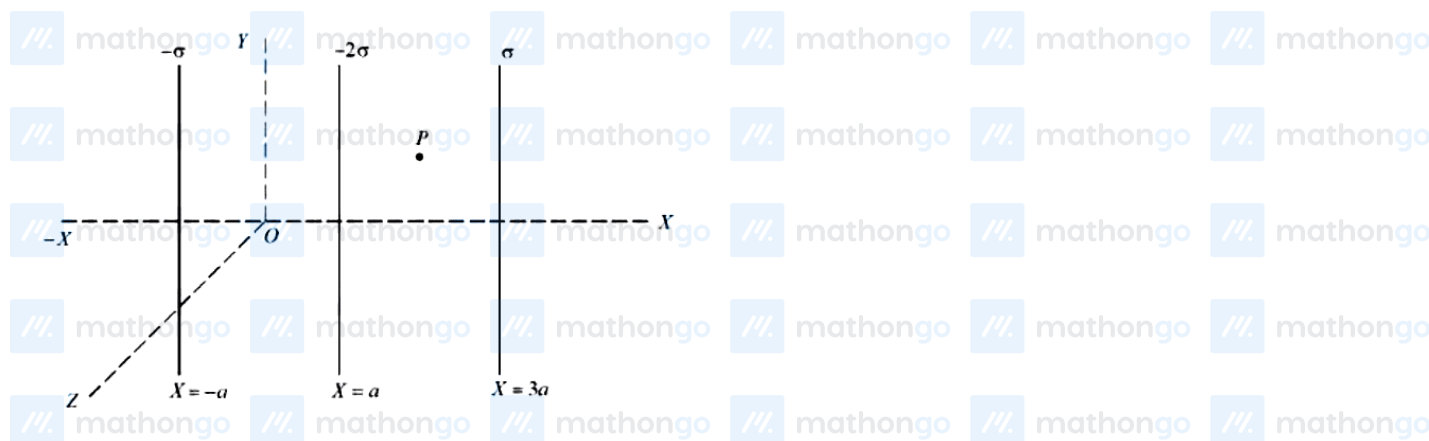
(4) $\sigma/2\epsilon_o$

Q7 - 2024 (06 Apr Shift 1)

Three infinitely long charged thin sheets are placed as shown in figure. The magnitude of electric field at the point P is $\frac{x\sigma}{\epsilon_o}$. The value of x is _____ (all quantities are measured in SI units).

Questions

MathonGo



Q8 - 2024 (06 Apr Shift 2)

Two identical conducting spheres P and S with charge Q on each, repel each other with a force 16 N. A third identical uncharged conducting sphere R is successively brought in contact with the two spheres. The new force of repulsion between P and S is :

- (1) 1 N
- (2) 6 N
- (3) 12 N
- (4) 4 N

Q9 - 2024 (08 Apr Shift 1)

Two charged conducting spheres of radii a and b are connected to each other by a conducting wire. The ratio of charges of the two spheres respectively is:

- (1) $\frac{a}{b} v$
- (2) $\sqrt{ab}$
- (3) $\frac{b}{a}$
- (4) ab

Q10 - 2024 (08 Apr Shift 1)

Questions

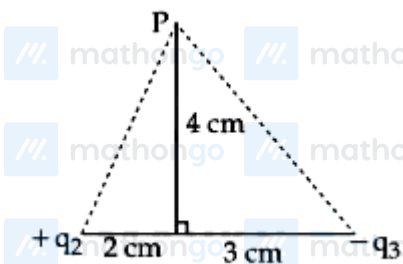
MathonGo

An electric field, $\vec{E} = \frac{2\hat{i}+6\hat{j}+8\hat{k}}{\sqrt{6}}$ passes through the surface of 4 m^2 area having unit vector $\hat{n} = \left(\frac{2\hat{i}+\hat{j}+\hat{k}}{\sqrt{6}}\right)$.

The electric flux for that surface is _____ Vm.

Q11 - 2024 (08 Apr Shift 2)

If the net electric field at point P along Y axis is zero, then the ratio of $\left|\frac{q_2}{q_3}\right|$ is $\frac{8}{5\sqrt{x}}$, where $x =$ _____.

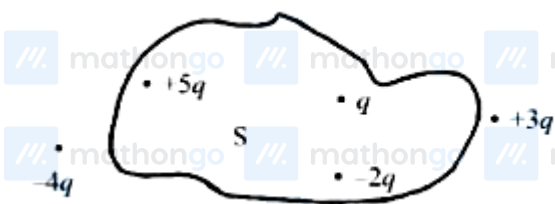


Q12 - 2024 (09 Apr Shift 1)

At the centre of a half ring of radius $R = 10 \text{ cm}$ and linear charge density 4 nCm^{-1} , the potential is $x\pi\text{V}$. The value of x is _____.

Q13 - 2024 (09 Apr Shift 2)

Five charges $+q$, $+5q$, $-2q$, $+3q$ and $-4q$ are situated as shown in the figure. The electric flux due to this configuration through the surface S is :



(1) $\frac{4q}{\epsilon_0}$

(2) $\frac{5q}{\epsilon_0}$

(3) $\frac{q}{\epsilon_0}$

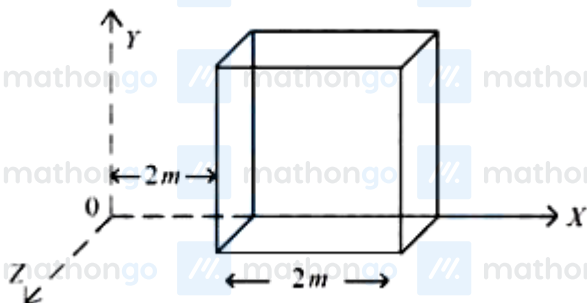
(4) $\frac{3q}{\epsilon_0}$

Questions

MathonGo

Q14 - 2024 (09 Apr Shift 2)

An electric field $\vec{E} = (2x\hat{i})NC^{-1}$ exists in space. A cube of side 2 m is placed in the space as per figure given below. The electric flux through the cube is _____ Nm^2/C .



Questions

MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

Answer Key

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

Q1 (3) MathonGo Q2 (16) MathonGo Q3 (1) MathonGo Q4 (4) MathonGo

Q5 (16) MathonGo Q6 (2) MathonGo Q7 (2) MathonGo Q8 (2) MathonGo

Q9 (1) MathonGo Q10 (12) MathonGo Q11 (5) MathonGo Q12 (36) MathonGo

Q13 (1) MathonGo Q14 (16) MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

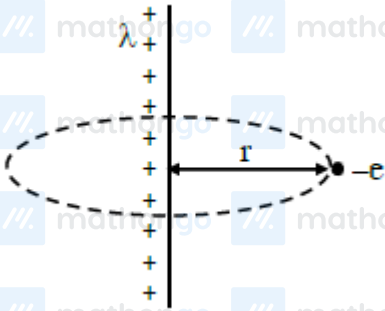
MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

#MathBoleTohMathonGo

Solutions

MathonGo

Q1



Electric field E at a distance r due to infinite long wire is $E = \frac{2k\lambda}{r}$

Force of electron $\Rightarrow F = eE$

$$F = e \left(\frac{2k\lambda}{r} \right)$$

$$F = \frac{2k\lambda e}{r}$$

This force will provide required centripetal force

$$F = \frac{m^2}{r} = \frac{2k\lambda e}{r}$$

$$v = \sqrt{\frac{2k\lambda e}{m}}$$

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}m \left(\frac{2k\lambda e}{m} \right)$$

$$= k\lambda e$$

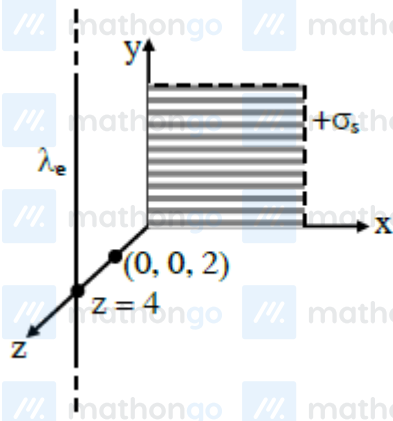
Q2

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

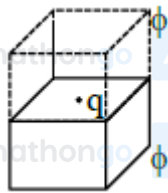
MathonGo



$$\begin{aligned} \frac{E_s}{E_\ell} &= \frac{\sigma}{2\epsilon_0} \times \frac{2\pi\epsilon_0 r}{\lambda} \\ &= \frac{\pi \times \sigma r}{\lambda} \\ &= \frac{\lambda}{\pi \times 2\lambda \times 2} = \frac{4\pi}{\lambda} \end{aligned}$$

$$\therefore n = 16$$

Q3



From

$$2\phi = \frac{q}{\epsilon_0}$$

$$\phi = \frac{q}{2\epsilon_0}$$

Q4

Static charge is developed due to air friction. This can result in combustion. So, metallic chains is used to discharge excess charge.

Q5

$$E_P = \frac{2KP}{r^3} = E$$

$$E_R = \frac{KP}{(2r)^3} = \frac{E}{16}$$

$$x = 16$$

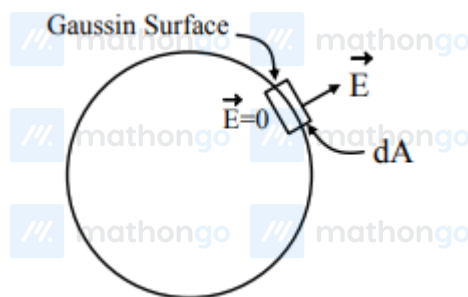
Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

Q6

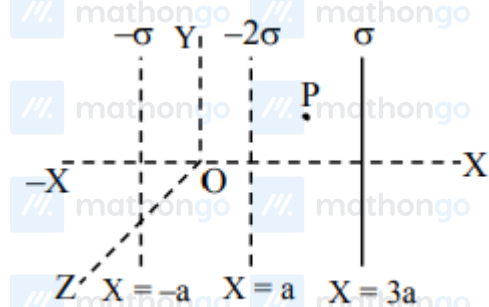


By Gauss law $\int \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$

$$EdA = \frac{\sigma \times dA}{\epsilon_0}$$

$$E = \frac{\sigma}{\epsilon_0}$$

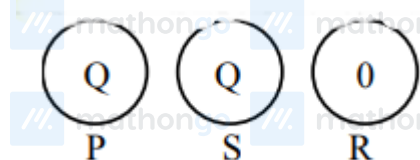
Q7



$$\vec{E}_P = \left(\frac{\sigma}{2\epsilon_0} + \frac{2\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} \right) (-\hat{i})$$

$$= -\frac{2\sigma}{\epsilon_0} \hat{i}$$

Q8



$$F_{PS} \propto Q^2$$

$$F_{PS} = 16 \text{ N}$$

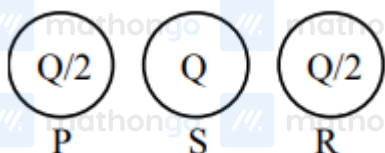
Now If P & R are brought in contact then

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

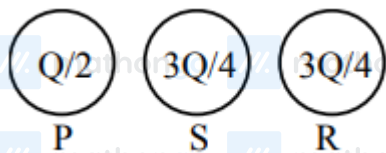
[Click here to download MARKS App](#)

Solutions

MathonGo



Now If S & R are brought in contact then



New force between P & S is :

$$F_{PS} \propto \frac{Q}{2} \times \frac{3Q}{4}$$

$$F_{PS} \propto \frac{3Q^2}{8} = \frac{3}{8} \times 16 = 6 \text{ N}$$

Q9

Potential at surface will be same

$$\frac{Kq_1}{a} = \frac{Kq_2}{b}$$

$$\frac{q_1}{q_2} = \frac{a}{b}$$

Q10

$$\phi = \vec{E} \cdot \vec{A}$$

$$= \left(\frac{2\hat{i} + 6\hat{j} + 8\hat{k}}{\sqrt{6}} \right) \cdot 4 \left(\frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}} \right)$$

$$= \frac{4}{6} \times (4 + 6 + 8) = 12 \text{ Vm}$$

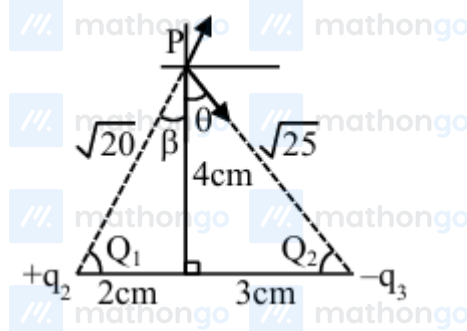
Q11

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo



$$\begin{aligned} \frac{Kq_2}{20} \cos \beta &= \frac{Kq_3}{25} \cos \theta \\ \frac{Kq_2}{20} \frac{4}{\sqrt{20}} &= \frac{Kq_3}{25} \frac{4}{\sqrt{25}} \\ \frac{q_2}{q_3} &= \frac{20}{25} \sqrt{\frac{20}{25}} = \frac{8}{5\sqrt{x}} \\ \Rightarrow \sqrt{x} &= \frac{8 \times 25\sqrt{25}}{5 \times 20\sqrt{20}} \\ x &= 5 \end{aligned}$$

Q12

Potential at centre of half ring

$$\begin{aligned} V &= \frac{KQ}{R} \\ V &= \frac{K\lambda\pi R}{R} \\ V &= K\lambda\pi \Rightarrow V = 9 \times 10^9 \times 4 \times 10^{-9} \pi \\ V &= 36\pi \end{aligned}$$

Q13

As per gauss theorem,

$$\begin{aligned} \phi &= \frac{q_{\text{in}}}{\epsilon_0} = \frac{q + (-2q) + 5q}{\epsilon_0} \\ \frac{4q}{\epsilon_0} \end{aligned}$$

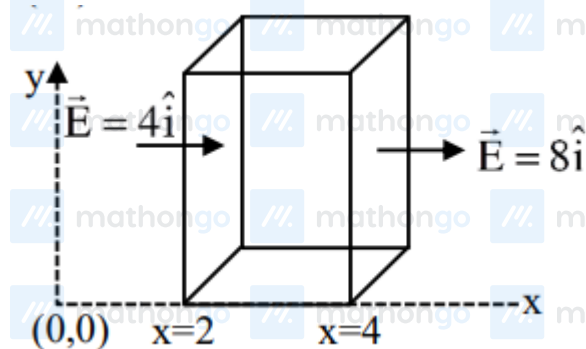
Q14

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo



$$\vec{E} = 2x\hat{i}$$

$$\phi = \vec{E} \cdot \vec{A}$$

$$\phi_{\text{in}} = -4 \times 4 = -16 \text{ Nm}^2/\text{c}$$

$$\phi_{\text{out}} = 8 \times 4 = 32 \text{ Nm}^2/\text{c}$$

$$\phi_{\text{net}} = \phi_{\text{in}} + \phi_{\text{out}} = -16 + 32 = 16 \text{ Nm}^2/\text{c}$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)